

# The angle at the centre

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Circle Theorems · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

Take two points **A** and **B** on a circle. Any angle standing on the arc they cut off **subtends** it. The **angle at the centre**,  $\angle AOB$ , is made by the two radii; an **angle at the circumference**,  $\angle APB$ , is made by two chords from a point **P** on the far arc. This lesson's keystone theorem: **the angle at the centre is exactly twice the angle at the circumference standing on the same arc** — so  $\angle AOB = 2 \times \angle APB$ . Read backwards, the circumference angle is half the centre angle. A striking consequence: every point **P** on that arc gives the *same*  $\angle APB$ , so all such angles are equal. The reason is a single extra line: join **P** to **O** and extend it, and two **isosceles** triangles (equal radii) do the rest.

## What you are learning to notice

- Angle at the **centre** =  $2 \times$  angle at the **circumference**, on the **same arc**.
- Read both ways: given the centre, **halve**; given the circumference, **double**.
- Every angle at the circumference on the same arc is **equal** (all are half of  $\angle AOB$ ).
- Always give the theorem as your **reason** — a circle-theorem answer needs a justification, not just a number.

## Moving through it, one step at a time

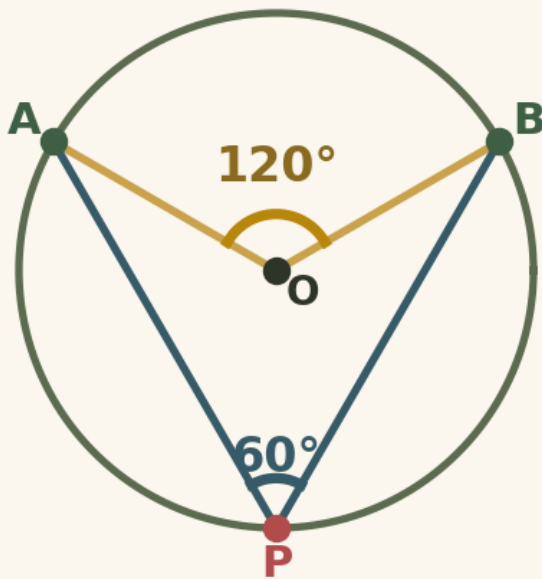
Identify the arc both angles stand on. Spot which angle you are given: if it is at the centre, halve it; if it is at the circumference, double it. Write the result and name the theorem as the reason. For the reflex case (**P** on the minor arc), the centre angle is the reflex one, and it is still twice the circumference angle.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

### Example 1 — Use the theorem both ways

O is the centre. The angle at the centre  $\angle AOB$  is  $120^\circ$ . Find the angle at the circumference  $\angle APB$ .

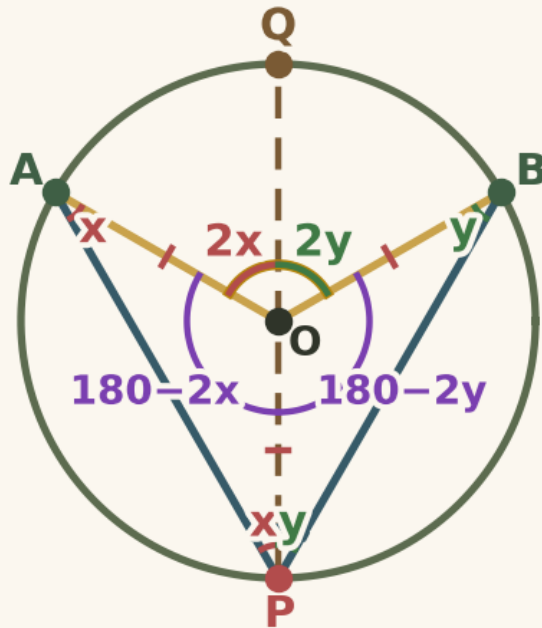


1. Both angles stand on the same arc AB.  $\angle AOB$  and  $\angle APB$  on arc AB
2. Angle at the centre = 2  $\times$  angle at the circumference.  $120 = 2 \times \angle APB$
3. So halve the centre angle.  $\angle APB = 120 \div 2$

**$\angle APB = 60^\circ$ , because the angle at the centre is twice the angle at the circumference on the same arc.**

### Example 2 — Why it is exactly double

Prove  $\angle AOB = 2 \times \angle APB$ . Join P to O and extend it to Q.



1.  $OA = OB = OP$  are radii, so triangles  $OAP$  and  $OBP$  are isosceles.

2. Isosceles  $\Rightarrow$  equal base angles:  $\angle OAP = \angle OPA = x$  and  $\angle OBP = \angle OPB = y$ .  $\angle APB = x + y$

3. Angles in each triangle sum to  $180^\circ$ , so the apex angles are  $\angle AOP = 180 - 2x$  and  $\angle BOP = 180 - 2y$ .

4. P, O, Q lie on a straight line, so  $\angle AOQ = 180 - (180 - 2x) = 2x$ , and  $\angle BOQ = 180 - (180 - 2y) = 2y$ .  $\angle AOB = 2x + 2y$

$\angle AOB = 2x + 2y = 2(x + y) = 2 \times \angle APB$ . The angle at the centre is twice the angle at the circumference. ■

### A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

### If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the parts of a circle and 'standing on an arc' (Lesson 1), angles in a triangle sum to  $180^\circ$ , and isosceles base angles are equal), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

#### How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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